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M.Tech AI DS

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## EDUCATION

### •Indian Institute of Information Technology Bhagalpur

2024 - 2026

M.Tech - AIML

CGPA/Percentage: 9.42 CGPA

### •Manipal Institute of Technology

2018 - 2022

B.Tech - Aeronautical Engineering

CGPA/Percentage: 7.02 CGPA

### •Delhi Public School, Jamshedpur, 12th

2018

Central Board of secondary Education

CGPA/Percentage: 77.4

### •Birla Public School, Doha, Qatar, 10th

2015

Central Board of Secondary Education - International (CBSE-i)

CGPA/Percentage: 9.6 CGPA

## TECHNICAL SKILLS AND INTERESTS

**Languages:** Python ,C++

**Developer Tools:** VS Code, Git, Docker, Hugging Face

**Frameworks:** LangChain, LangGraph, TensorFlow, PyTorch, QuantLib, Flask, FastAPI

**Cloud/Databases:** ChromaDB, PostgreSQL

**Soft Skills:** Problem Solving, Research Oriented Mindset and Analytical Thinking, Strong Communication and Highly Adaptable to Individual or Team Roles.

**Coursework:** DeepLearning.AI - Generative AI with Large Language Models

**Areas of Interest:** Large Language Models, Agentic Systems, AI Agents

## PROJECTS

### •Predictive Maintenance

January 2025 - May 2025

*Forecasting Remaining Useful Life of Turbo-Fan engines under different operating conditions.*

- Preprocessed sensor data using Kalman Filters for denoising and Discrete Wavelet Transform (DWT) for feature extraction.
- Developed LSTM-based models (standard, Bi-LSTM, and attention variants) to predict Remaining Useful Life (RUL) scores.
- Achieved competitive RMSE and MAE scores compared to existing hybrid LSTM approaches for Remaining Useful Life (RUL) estimation.

### •Explainable AI for UAV Crop Diagnostics

January 2026 - May 2026

*Automated monitoring and diagnostic analysis of maize crop diseases using UAV imagery and XAI*

- Designed and implemented a custom, lightweight Convolutional Neural Network (CNN) optimized for plant pathology, achieving a 97.06% test accuracy across a dataset of 5,122 high-resolution UAV images.
- Engineered a novel "GI Fusion" Explainable AI (XAI) framework by mathematically synthesizing Grad-CAM's regional localization with the pixel-level precision of Integrated Gradients.
- Reduced the model's footprint to under 1 million parameters (0.82M), enabling a rapid inference latency of 8.2 ms per image for real-time drone edge-deployment.
- Created transparent, verifiable attribution maps that successfully isolated true biological symptoms of Maydis Leaf Blight from background environmental noise and weed mimicry.

### •Autonomous Multi-Agent Research Assistant

April 2026 - Present

*Agentic workflow for automated academic literature retrieval, evaluation, and synthesis.*

- Developed a cyclical multi-agent architecture using LangGraph to autonomously route complex queries, utilizing tool-calling for web search and API data extraction.
- Implemented an advanced Retrieval-Augmented Generation (RAG) pipeline, embedding academic documents into a ChromaDB vector store for semantic long-term memory.
- Engineered self-correcting logic loops within the agent state graph to evaluate retrieved context relevance, effectively mitigating LLM hallucinations during technical report generation.
- Automated the synthesis of comparative literature reviews on Deep Learning architectures and Explainable AI (XAI) methodologies, outputting structured, human-readable markdown reports.